

Investigation 8.4: Back-emf of an electric motor

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Background

A small D.C. electric motor that uses a permanent magnet to produce the magnetic field generates a back-emf when its armature rotates. This back-emf opposes the voltage applied to the motor and reduces the current through the motor. If the motor stops because of a heavy load being applied to it the current will increase because there is no back-emf.

Pre-lab

A researcher applies a fixed voltage of 12 V DC to a small permanent magnet DC electric motor. By varying the load on the motor shaft she was able to reduce the speed of the motor and measure the current at various speeds. A fixed voltage of 12 V was applied to the motor. Some of her results are tabulated here:

Current (I) drawn by motor (mA)	550	515	270	140	0
Motor speed (S) revolutions per minute (rpm)	0	200	400	600	800

- Plot a graph of the current drawn (I) versus the motor speed (S).
 - What does this graph suggest about the relationship between I and S?
 - Determine the mathematical equation relating I and S.
- Estimate the resistance of the motor's armature winding. (Hint: There is no back-emf when the motor is stalled.)
- If the back-emf (V) of the motor is related to the speed S by the formula $S = kV$ determine the value of the constant k.

The Task

Use a small motor and appropriate apparatus to create your own set of results. Be aware that applying too high a voltage, or running a current through a stalled motor, can damage the motor beyond repair.

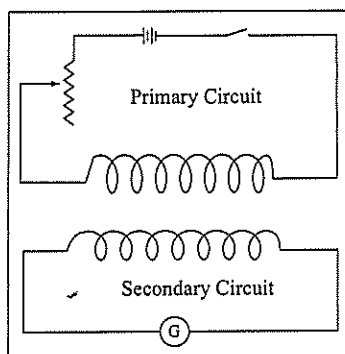
In what ways were your results similar to, and different from, the ones shown in the pre-lab? Why can running current through a stalled motor cause damage? Explain why a motor produces a back-emf when operating but not when the motor stalls.

Notes

Problem Solving and Calculations

Set 7: Magnetic Induction

Notes



1. An aircraft flies at 980 km h^{-1} from west to east at an altitude of 10 km. If the vertical component of the Earth's magnetic field is $3.5 \times 10^{-5} \text{ T}$ and the aircraft has a wingspan of 60 m, calculate the potential difference it develops between its wing tips.
2. A portable 50 W television set is designed to operate on a 12 V battery, or by using a mains supply of 240 V connected through a transformer.
 - a) If the transformer in the set is 90% efficient, what current does the television set draw if it is operating on mains power?
 - b) Explain why this television set has vents in the walls of its casing.
3. A physics student inserted the north pole of a bar magnet into a coil that was connected to a galvanometer, and noticed that the galvanometer needle moved to one side. When she withdrew the magnet from the coil she notice the needle moved to the opposite side of the galvanometer. Explain these observations with the help of a diagram.
4. Referring to the diagram at left
 - a) explain why, when you close the switch in the primary circuit, you detect a transitory electric current in the secondary circuit;
 - b) suggest at least two ways in which you can increase induced voltage in the secondary circuit; and
 - c) explain in which direction the current flows in the secondary circuit relative to the primary circuit.
5. A 500 mm long radio aerial is attached to, and insulated from, the roof of a taxi. If the taxi is moving in an easterly direction at 60 km h^{-1} and the horizontal component of Earth's magnetic field is $2.5 \times 10^{-5} \text{ T}$, calculate:
 - a) the emf induced in the aerial;
 - b) the rate at which the aerial cuts the magnetic flux.
 - c) If the taxi turns a corner maintaining the same speed but now travels in a southerly direction, does the induced emf remain the same? Explain.
6. An electrical engineer working on the design of a new electric train said: 'If the opposite of Lenz's law was true, the motors in this train would soon burn out.' Explain why this would happen.
7. A generator coil of 200 turns is 30 mm in diameter. It experiences a uniform flux density change of 0.5 T in 10 s. Calculate the average emf induced in the coil.
8. A student connected the ends of a circular loop of wire with a radius of 0.10 m to a 5.0Ω resistor. He put the coil near a transformer that, at a particular instant directed a magnetic field of 0.25 T at right angles to the plane of the coil.
 - a) Calculate the emf induced in the coil if the magnetic field dropped to zero in 0.20 s.
 - b) Draw a diagram of the coil and the resistor and show the direction of the magnetic field passing through the coil as it dropped to zero.
 - c) Indicate the direction of the induced current through the resistor.
 - d) Calculate the value of the induced current through the resistor.

Problem Solving and Calculations

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Set 7: Magnetic Induction

9. A train is travelling with a constant velocity of 80 km h^{-1} in an area where the vertical component of the Earth's magnetic field is $36 \mu\text{T}$.
 - a) What is the size of the emf induced across each 1.0 m long axle?
 - b) If the train is travelling in a south westerly direction, describe the force acting on an electron in this axle.
10. A student timed the oscillation of a pendulum made of an aluminium plate. She noticed the period of oscillation was much less when she made the pendulum plate swing between the poles of a strong horseshoe magnet. Explain this observation. (Note that if you look at the electric power meter in your home you should see an aluminium disc rotating through the poles of a strong magnet.)
11. A commercial AC generator operates at a frequency of 50 Hz and produces a maximum voltage of 180 V . If the area of the coil is $2.0 \times 10^{-2} \text{ m}^2$ and the armature rotates in a magnetic field of strength 0.20 T , how many turns of wire must the coil have to produce the maximum voltage? Assume the maximum flux through the coil reduces uniformly to zero over one quarter of a rotation.
12. In a laboratory investigation a student moved 20 mm of a length of copper wire perpendicularly across a 0.50 T magnetic field. The ammeter connected to the ends of the wire indicated a current of 10 mA .
If the total resistance of the circuit was 2.5Ω , calculate:
 - a) the constant velocity with which she moved the wire through the field,
 - b) the force she exerted to maintain this constant velocity.
13. A simple generator contains a square armature coil of side lengths 200 mm by 200 mm . The coil contains 300 turns of copper wire. It rotates at 60 revolutions per second in a uniform magnetic field
 - a) What is the strength of this field if the generator produces an average voltage of 240 V ?
 - b) At what point in its rotation is the peak voltage produced? At what point is the voltage zero.

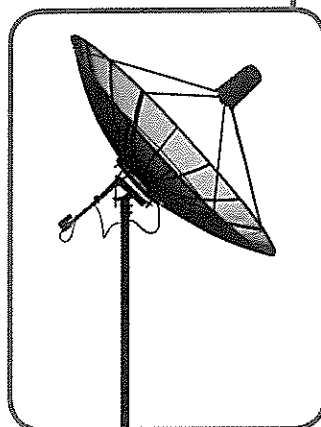


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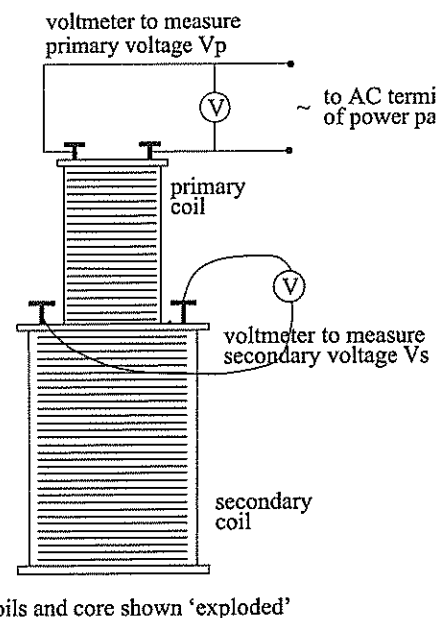
Problem Solving and Calculations

Set 8: Electrical Energy and Power

Notes



1. A defibrillator is a device medical personnel use to return an irregular heart beat to its normal rhythm. A defibrillator passes a 20 A current at 3000 V through a patient's heart in about 5 ms.
 - a) How much electrical power does a defibrillator provide?
 - b) How much electrical energy passes through a patient?
 - c) What is the resistance of the patient's body in this circuit?
2. A particular communications satellite consumes electrical energy at a rate of 2000 J s^{-1} . The satellite is powered by a solar panel that receives 1373 J of per second from the Sun on each square metre of panel.
 - a) If the panel converts solar energy into electrical energy at 10.0% efficiency, what is the area of the panel required to meet the satellites energy needs?
 - b) If the panel provides an output voltage of 50.0 V under optimum load, calculate the load's resistance.
3. An electric train has a motor rated at 1.50 kV, 125 kW.
 - a) Calculate the current the motor draws.
 - b) Calculate the circuit resistance.
4. While performing a laboratory investigation on the structure of a step-down transformer, a student noticed that the wire in one of the coils was thicker than the wire in the other coil. Does the thicker coil belong to the primary coil or the secondary coil? Explain why this is so.
5. The diagram at right shows a simplified view of a transformer without its metal core plates. It consists of two coils of wire, one inside the other. If the current in the outer coil is flowing in an anticlockwise direction and is increasing, explain how you would determine the direction of the current induced in the inner coil. Lenz's Law may be useful in explaining your answer.
6. A transformer in a neon sign changes 240 V to 12 kV.
 - a) Assuming that the transformer is ideal, compare the magnitude of the current in the secondary coil to the magnitude of the current in the primary coil.
 - b) If the primary coil had 200 turns of wire, how many turns should there be in the secondary coil?
 - c) If the transformer is actually 98% efficient, how large is the current in the secondary coil compared with the current in the primary coil?



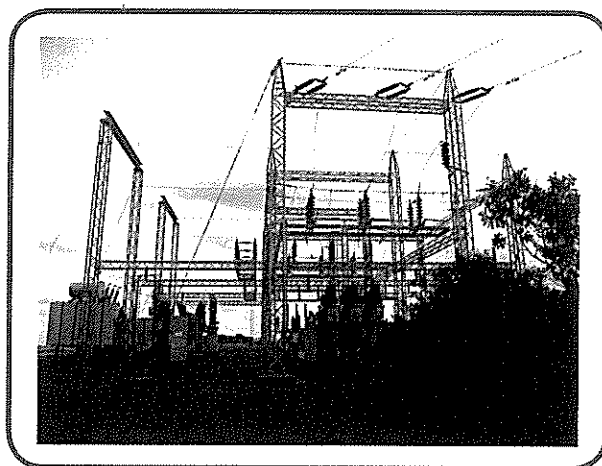
Problem Solving and Calculations

Set 8: Electrical Energy and Power

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Notes

7. An electric jug has a rating of 240 V, 2.50 kW.
 - a) Calculate the jug's resistance.
 - b) If the jug is used to heat water over a 2.00 minute period, how much electrical charge will have passed through the jug in that time?
8. A car engine needs 1.00 kW to start it. The starter motor supplies this power. The starter motor is 80% efficient and it runs off a 12.0 V battery.
 - a) Calculate the current the starter motor draws from the battery to start the engine.
 - b) In what way is the diameter of the copper wire that joins the battery to the starter motor quite different to the other electrical wires in the car? Why is this so?
9.
 - a) For a typical electricity grid system, what is the role of a 'sub-station'?
 - b) Suggest some reasons why an electricity company would not use the same sub-station to supply electricity to both an electric train system and to nearby houses.
 - c) A power station generator produces a voltage of 18.0 kV, but the voltage in the transmission line that carries the electrical energy to the suburban sub-stations is 330 kV. Why does the electrical utility 'step up' the voltage before it is transmitted?



A substation

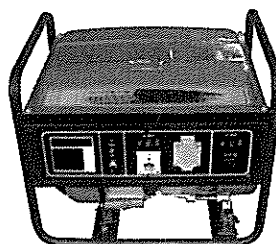
10. An electricity generation plant in a small town produces an average of 500 kW of electric power. The 10 km length transmission line from the power plant to the town's sub-station has a total resistance of $0.50\ \Omega$. The voltage is stepped up at the plant from 2.0 kV to 20 kV and then down to 240 V in the town.
 - a) If the plant had transmitted electricity at 2.0 kV, what power loss would have occurred in the transmission line when working at the average power output?
 - b) Under normal circumstances the plant transmits at 20.0 kV. What power loss would occur in the transmission line at this voltage?
 - c) Compare your answers to a and b. Give a reason why power is usually transmitted at very high voltages.
11. Electric utilities usually transmit energy at very high voltages. However, engineers have worked out that voltages over 1000 kV are uneconomical, and have an impact on the environment. Why should this be so?

Problem Solving and Calculations

Set 8: Electrical Energy and Power

Notes

12. An electric motor in a goods lift needs a minimum voltage of 405 V to operate. The cable supplying power to the motor comes from a transformer that has an output of 415 V. When the lift is operating, the cable carries a current of 200 A and has a resistance of $4.0 \times 10^{-1} \Omega$ per metre of cable.
- Determine the power loss in the cable when the voltage at the electric motor drops to 405 V.
 - Determine the maximum length of cable that can supply the minimum of 405 V to the motor.
 - How would you change this arrangement to allow the motor to work at a larger distance from the transformer?
13. A portable generator can provide 5.0 kW of electrical power. A petrol engine drives the generator. The generator is 80% efficient.
- How much mechanical power must the engine provide?
 - Explain where you think the power losses occur in the generator.



14. Electric motors, fluorescent lighting systems and arc welding equipment all use some of the electrical energy they consume to create magnetic fields. If you have several such devices operating at your house, what effect would this have:
- on the voltage available to other devices on your property?
 - on the temperature of the supply lines to your property?
 - on the brightness of electric lights on your property?
15. A 3.0 MW substation provides current at 25 kV to electric trains through overhead wires. The wires are designed to operate in the voltage range 20 to 25 kV. If the resistance of the overhead wires is $1.2 \Omega \text{ km}^{-1}$, how far from the substation can the train travel before the supply voltage drops below its operating needs?

- 14 a) 383 mm from the balcony
 15 $1.86 \times 10^5 \text{ N}$; $1.95 \times 10^5 \text{ N}$
 20 7.76 kg; 2.57 m from the T_1 end.
 21 60 cm
 23 a) 137 N
 b) lower hinge 172 N toward the door at 36.9° above the horizontal
 upper hinge 172 N toward the wall at 36.9° above the horizontal
 24 a) $1.95 \times 10^3 \text{ N}$
 b) vertical = 109 N upward, horizontal = $1.83 \times 10^3 \text{ N}$ out from wall
 c) $1.84 \times 10^3 \text{ N}$ out from wall at 3.40° above the horizontal
 25 tension = $1.09 \times 10^3 \text{ N}$; force at hinge = 551 N out from the wall and 10.1° below the horizontal
 26 2.07 m from P
 27 a) 156 N
 b) vertical component = 147 N, horizontal component = 156 N
 c) 214 N to right and 43.3° below the horizontal
 28 a) $2.39 \times 10^5 \text{ N}$
 b) $2.97 \times 10^5 \text{ N}$
 29 a) 18.4 kg
 b) 144 N at 26.9° above the horizontal

Electricity and Magnetism

- Set 6 5 0.042 N m
 10 2.4 N
 12 0.075 N downward
 Set 7 1 0.57 V
 2 a) 2.3 A
 5 a) 0.21 mV
 5 b) $2.1 \times 10^{-4} \text{ Wb s}^{-1}$
 7 7.1 mV
 8 a) 39 mV
 8 d) 7.9 mA
 9 a) $8.0 \times 10^{-4} \text{ V}$
 11 225 turns
 12 a) 2.5 m s^{-1}
 12 b) $1.0 \times 10^{-4} \text{ N}$
 13 a) 83 mT
 Set 8 1 a) 60 kW
 1 b) 300 J
 1 c) 150Ω
 2 a) 14.6 m^2
 2 b) 1.25Ω
 3 a) 83 A
 3 b) 18Ω
 6 a) 0.0200 times
 6 b) 10 000 turns
 6 c) 0.0196 times
 7 a) 23Ω
 7 b) $1.3 \times 10^3 \text{ C}$
 8 a) 100 A
 10 a) 31.3 kW
 10 b) 0.31 kW
 12 a) 2.00 kW
 12 b) 0.125 m
 13 a) 6.25 kW
 15 35 km